

ANNA MANUKYAN

Applied Mathematics | Machine Learning | Computer Vision

@anmanukyan0801@gmail.com

+374 95 707042

Yerevan, Armenia

ML & RESEARCH EXPERIENCE

Dental X-ray Analysis: Segmentation & Detection

BSc Thesis — Lomonosov Moscow State University

2025

- Developed a two-stage deep learning pipeline for analysis of panoramic dental X-rays using **PyTorch**.
- Trained a **U-Net with ResNet34 encoder** for tooth-region segmentation, achieving $\text{Dice} \approx 0.96$ and $\text{IoU} \approx 0.92$.
- Built a **Faster R-CNN with ResNet50 + FPN** to detect cavities, fillings, implants, and impacted teeth, achieving $\text{mAP@0.5} \approx 0.79$.
- Prepared a segmentation dataset of **1,600 panoramic X-rays**, including **600 images manually annotated** for the project.

Machine Learning Intern

Usetech

2024

- Developed a computer vision pipeline for detecting **text blocks and text lines** in Arabic handwritten manuscript images using **YOLOv8**.
- Prepared annotated image data and implemented preprocessing and conversion of XML annotations into YOLO-compatible formats.
- Worked with standard object detection and **oriented bounding box (OBB)** models.
- Contributed to automated model training and inference workflows.
- Project Repository

EDUCATION

B.Sc. in Applied Mathematics and Informatics

Lomonosov Moscow State University

2021 – 2025

GPA: 4.42 / 5.0

M.Sc. in Computer & Information Science

American University of Armenia

2026 – Present

PROFESSIONAL EXPERIENCE

Project Manager

Weisseco LLC

2023 – 2025

- Managed financial and document flows and tracked project income and expenditures.
- Prepared commercial proposals and participated in negotiations.
- Adapted and optimized business processes in response to changing project requirements.

TECHNICAL SKILLS

Programming

Python Java C/C++

Machine Learning & Computer Vision

PyTorch YOLOv8 / Ultralytics
U-Net ResNet
Faster R-CNN FPN

Data & Image Processing

NumPy Pandas
Matplotlib Albumentations
OpenCV

ML Methods

Semantic Segmentation
Object Detection
Transfer Learning
Data Augmentation
Oriented Bounding Boxes

MATHEMATICAL FOUNDATIONS

Linear Algebra

Calculus

Probability & Statistics

Discrete Mathematics

Differential Equations

Mathematical Analysis

Equations of Mathematical Physics

COMPUTER SCIENCE

Data Structures

Algorithms

Object-Oriented Programming

Computer Science Fundamentals

ADDITIONAL EDUCATION

Neural Networks & Computer Vision

Stepik

CS50: Introduction to Computer Science

Database Fundamentals

Teach-In

LANGUAGES

Armenian — Native

English — Fluent

Russian — Fluent